Semantic Image Clustering Using Relevance Feedback

Xiaoxin Yin
Department of Computer Science
Tsinghua University, Beijing 100084, China
yinxx@hotmail.com

Mingjing Li, Hong-Jiang Zhang
Microsoft Research China
49 Zhichun Road, Beijing 100080, China
{mjli, hjzhang}@microsoft.com

Abstract

This paper describes an image clustering approach to grouping semantically similar images. In this approach, the similarity between images is estimated using users' relevance feedback information recorded in the user log of an image retrieval system. An algorithm similar to CAST (Cluster Affinity Search Technique) is used to identify clusters of semantically related images. It is a two-stage clustering method: the pre-classification partitions the images into closely related groups; within each group, the fine clustering mines semantically related clusters of images. Experiments on more than 10,000 images demonstrate the effectiveness of this approach.

Keywords

Image retrieval, relevance feedback, clustering, semantic, annotation

Semantic Image Clustering Using Relevance Feedback*

Xiaoxin Yin
Department of Computer Science
Tsinghua University, Beijing 100084, China
yinxx@hotmail.com

Mingjing Li, Hong-Jiang Zhang
Microsoft Research China
49 Zhichun Road, Beijing 100080, China
{mjli, hjzhang}@microsoft.com

Abstract

This paper describes an image clustering approach to grouping semantically similar images. In this approach, the similarity between images is estimated using users' relevance feedback information recorded in the user log of an image retrieval system. An algorithm similar to CAST (Cluster Affinity Search Technique) is used to identify clusters of semantically related images. It is a two-stage clustering method: the pre-classification partitions the images into closely related groups; within each group, the fine clustering mines semantically related clusters of images. Experiments on more than 10,000 images demonstrate the effectiveness of this approach.

1. Introduction

Content-based image retrieval (CBIR) has drawn widespread research interests. However, most CBIR systems rely on low-level image features, such as color and texture, to find similar images. Unfortunately, the performance of CBIR systems is usually very poor because the low-level features cannot capture the semantic content of images. On the other hand, relevance feedback can improve the image retrieval performance, and many techniques have been proposed [3,6,8]. In a relevance feedback retrieval system, the user labels one or more image examples that are relevant or irrelevant to the query in the retrieval results. The feedback information reveals the semantic relations among images.

Another approach to image retrieval is based on keyword annotation, in which human operators assign keywords to images, and text-based retrieval techniques are utilized to search for images. Obviously, keyword annotations represent the semantic content of an image more precisely than low-level image features. However, image annotation is a tedious process. Thus, it is practically impossible to manually annotate all images on the Internet.

In this paper, we propose an image clustering approach that can relate relevance feedback with image annotation. Unlike the semantic network [5], which is built on the top of keyword association with images, our method is based on clustering images into semantically related groups, which can be used for further categorization and semi-automatic annotation. For example, keywords may be manually assigned to image clusters and then automatically propagated into individual images. In this way, human effort can be reduced significantly.

In our approach, an image similarity matrix is built using the relevance feedback information collected in the user log of an image retrieval system. Based on this matrix, a two-stage clustering is performed to classify images into closely related clusters, such that each cluster represents a particular semantic meaning. The basic clustering algorithm is similar to the cluster affinity search technique [2], except that the affinity of an image to a cluster is redefined, and a different criterion is adopted to construct a cluster. During the clustering process, the number of clusters is automatically determined by the algorithm and the data available; and an image may belong to more than one cluster, in order to reflect the fact that one image may have several semantic meanings.

The remainder of this paper is organized as follows. We discuss previous work related to this paper in Section 2. We define the image similarity matrix for clustering in Section 3, and describe the clustering algorithm in Section 4, respectively. In section 5 we introduce a potential application for semi-automatic annotation. In Section 6 we evaluate our approach on a database of more than 10,000 images and compare it with graph-theoretic clustering method. Finally, we conclude in Section 7.

2. Related Works

Clustering can be considered the process of organizing data in the input set into subsets whose members are similar in some way. It has been studied extensively in many different areas, such as pattern recognition, speech

* This work was performed at Microsoft Research China.

and image processing, computational biology, and so forth. Pair-wise data clustering is a combinational optimization problem, in which the data is indirectly characterized by pair-wise comparisons. In our particular application, the similarity between two images is estimated from relevance feedback, and the goal is to extract semantic classes of images. So it is a specific application of the pair-wise data clustering algorithms.

Through experiments, we find that some existing algorithms do not deal with our problem very well. Both simulated and deterministic annealing algorithms [4] are computationally extensive, and it is difficult to determine the number of clusters. Density-based methods [7] result in large clusters, which are difficult to interpret. The graph-theoretic clustering method [1] does not utilize the similarity information.

Our clustering approach is similar to the cluster affinity search technique, which is called CAST [2]. That algorithm requires no prior knowledge of the number of clusters. It operates by constructing one cluster at a time. The input is a pair (S,t), where S is a symmetric $n \times n$ similarity matrix, and t is the affinity threshold. The affinity of an element x with respect to a cluster C is defined as $a(x) = \sum_{y \in C} S(x,y)$. An element x has high affinity if $a(x) \geq t|C|$; otherwise, it has low affinity. The algorithm alternates between adding high affinity elements to C and removing low affinity elements from it. When C is stable, it is saved as a cluster. This process iterates until all elements have been assigned to a cluster.

In order to better fit the problem, some modifications are made to the above algorithm. The clustering process is organized into two stages for fast processing of large data set. In the pre-classification stage, a different affinity is defined. In the fine clustering stage, a different criterion is used to build a cluster.

3. Semantic Similarity between Images

As stated above, the similarity between two images is defined based on the relevance feedback information recorded in the user log of an image retrieval system. The basic assumption is that if two images are labeled as positive examples in a search session, it is very likely that they contain similar semantic content. And the higher the number of sessions in which they are positive examples, the higher the semantic similarity between them may be expected. Based on this assumption, the similarity between two images is defined as $s_{xy} = (c_{xy}/M)^{\theta}$ $(0 < \theta \leq 1)$, where c_{xy} is the number of search sessions in which both image x and y are labeled as positive examples, while M is defined as $M = 1 + \max(c_{xy})$. θ is introduced as a smoothing factor in order to smooth the raw data, because c_{xy} is usually very small in our data collection. θ is set to 0.5 in the experiments.

Based on the image similarities, a similarity graph is constructed, in which a node represents an image, and an edge exists between two nodes if the similarity between the corresponding images is above zero. This graph fits the corrupted clique model [2] very well. Because of the limited data collected in the user log, the similarity between two images may be very small or zero even if they actually contain the same semantic content; on the other hand, the similarity may be high when two images are semantically unrelated to each other, due to users' search diversity and noise data. This is one of the reasons we choose a CAST-like algorithm to mine semantic clusters of images.

4. Semantic Image Clustering

Our image clustering approach is a two-stage method. In the pre-classification stage, each image is assigned into one and only one group. In the fine clustering stage, an image within each group may be assigned to more than one cluster, reflecting the fact that one image may have several semantic meanings. And an image may also not belong to any cluster, because of the limited data available. The first stage is used to deal with large data set efficiently. However, if the amount of data is not large, the first stage may be skipped, and the second stage can be applied to the whole data set directly.

4.1. Definitions

The neighborhood of an image x with respect to a data set T is defined as $N(x,T) = \{y \mid y \in T, s_{xy} > 0, y \neq x\}$.

The relation of an image x with a data set T is defined as $R(x,T) = \sum_{y \in N(x,T)} s_{xy}$.

A cluster C is a subset of images from the whole image set S.

The affinity of an image x to a cluster C is defined as

$$A(x,C) = \left(\frac{R(x,C)}{|C|}\right)^{\alpha} \left(\frac{R(x,C)}{R(x,S)}\right)^{\beta},$$

where α and β are control factors. Both are set to 0.5 in our experiments. This definition takes account of the relations between image x and images both inside and outside cluster C.

The density of a cluster C is defined as

$$D(C) = \frac{\sum_{x \in C} R(x,C)}{|C|^2} \left(1 - \frac{1}{|C|}\right)^{\gamma},$$

where the left part of the definition is the average similarity among images in cluster C, and the right part is

used to control the size of C in the clustering process. γ is set to 3 in our experiments.

4.2. Image Classification

The pre-classification algorithm is similar to CAST except a different definition of affinity and a different cluster initialization. At first, the image with the maximum relation to the unclassified image set is identified. Then a class is formed with this image and its neighbors. After that, this class is iteratively refined by adding images with strong affinities to the class and removing images with weak affinities from it. The algorithm is described as follows.

Set U *to the whole image set*
Set ε *to the affinity threshold*
WHILE $U \neq 0$ *DO*
  $m = \arg\max_x R(x,U)$
  $C = N(m,U) \bigcup \{m\}$
  REPEAT
    $C' = C$
    $C = \{x \mid A(x,C') \geq \varepsilon, x \in U\}$
  WHILE $C \neq C'$
  Save C *as a class*
  $U \leftarrow U - C$
END

Like CAST, this algorithm constructs one class at a time. Meanwhile, several images may be added to or removed from a class in a single iteration. The resulting class may contain images with different semantic meanings. The algorithm described below is able to identify small image clusters with homogeneous semantic meanings within a class.

4.3. Image Clustering

The fine clustering algorithm is also similar to CAST. However, the criterion to construct a cluster is different. An image is added to or removed from a cluster such that the overall density of this cluster is improved, no matter whether the affinity of this image is above or below a threshold.

Initially, an image is selected as seed from an image set or a class built by the above classification algorithm, and a cluster is constructed with this seed. Then the cluster is iteratively updated until its density is maximized. The seed does not necessarily remain in the resulting cluster.

Since every image in the data set may be a seed, it seems that the number of expected clusters is equal to the size of the data set. In fact, many resulting clusters are identical to each other, and some clusters have heavy overlaps with others. So some heuristic rules are used to prune redundant clusters. For identical clusters, only one is kept; for clusters with heavy overlaps, the one with the highest density is reserved and all others are removed. Clusters with few images are also pruned. After post processing, only a few clusters are left, which contain semantically homogeneous images.

The algorithm is illustrated below.

FOR each image x *in class* C
  $C_x = \{x\}$
  REPEAT
    FOR each image y *in class* C
      IF $y \in C_x$ *THEN*
        $C_y = C_x - \{y\}$
      ELSE
        $C_y = C_x \bigcup \{y\}$
      END IF
    END FOR
    $m = \arg\max_y D(C_y)$
    IF $D(C_m) > D(C_x)$ *THEN*
      $C_x = C_m$
    END IF
  UNTIL no change in C_x
  Save C_x *as a cluster*
END FOR

In this way, an image may be assigned to more than one cluster. This conforms to the observation that one image may convey several semantic meanings. An image may also not belong to any cluster, due to the limited data available.

5. Application in Semi-Automatic Annotation

Keyword annotation is of great help to image retrieval. However, it is impractical for large image database. The proposed semantic image clustering approach may provide a solution to this problem. After clustering, a cluster can be manually annotated according to its semantic meaning. Then the annotation can be automatically propagated into an image, with a weight proportional to its affinity to this cluster.

6. Experimental Results

We implemented the algorithm and evaluated it on a database of more than 100,000 images collected from the Internet. Among these images, only 12,509 were labeled by users' relevance feedback. So the experiments are conducted on this subset of images.

In the first stage, images are classified into 304 classes, from which more clusters are extracted in the second stage. After post processing, only 395 clusters are resulted, which cover 10,351 images in total. Some

images do not appear in any cluster, while some appear in several clusters.

Image examples from cluster “mountain” are given in Figure 1, while examples from “lake” in Figure 2. Please note that two images appear in both clusters, because they contain both “mountain” and “lake”.

Figure 3, 4, and 5 present more image examples. Images in each cluster are really homogeneous in semantics.

We also implemented the graph-theoretic clustering method [1] and applied it to the same database. A total of 296 clusters are constructed, which cover 11,327 images. We manually checked images in each cluster, and found that clusters are less homogeneous than the results of our approach. According to the definition of the cluster “compactness” [1], the average “compactness” is 0.9616, while that by our approach is 0.9927. This reveals that our approach is more appropriate to this specific problem.

7. Conclusion

Based on relevance feedback, the proposed image clustering approach is able to extract semantically homogeneous clusters of images. It may be helpful in semi-automatic annotation and categorization for image retrieval. Experiments show that it is superior to other clustering algorithms for this particular application.

8. Acknowledgements

We would like to thank Fang Qian and Hui Yu for providing technical comments.

9. References

[1] S. Aksoy and R.M. Haralick, “Graph-Theoretic Clustering for Image Grouping and Retrieval”, in Proc. IEEE CVPR, 1999.

[2] A. Ben-Dor, R. Shamir, and Z. Yakhini, “Clustering Gene Expression Patterns”, Journal of Computational Biology, 1999.

[3] I.J. Cox, M.L. Miller, S.M. Omohundro, and P.N. Yianilos, “PicHunter: Bayesian Relevance Feedback for Image Retrieval”, in Proc. Int. Conf. on Pattern Recognition, 1996.

[4] T. Hofmann and J.M. Buhmann, “Pairwise Data Clustering by Deterministic Annealing”, in IEEE Trans. on Pattern Analysis and Machine Intelligence, Vol.19, No. 1, January 1997.

[5] Y. Lu et al, “A Unified Framework for Semantics and Feature Based Relevance Feedback in Image Retrieval Systems”, in Proc. ACM Multimedia, 2000.

[6] Y. Rui, T.S. Huang, and S. Mehrotra, “Relevance Feedback Techniques in Interactive Content-Based Image Retrieval”, in Proc. IS&T and SPIE Storage and Retrieval of Image and Video Databases, 1998.

[7] Z. Su et al, “Correlation-based Document Clustering using Web Logs”, in Proc. 34th Hawaii Int. Conf. on System Sciences (HICSS-34), Maui, Hawaii, January 2001.

[8] N. Vasconcelos and A. Lippman, “Learning from User Feedback in Image Retrieval Systems”, NIPS’99, Denver, Colorado, 1999.

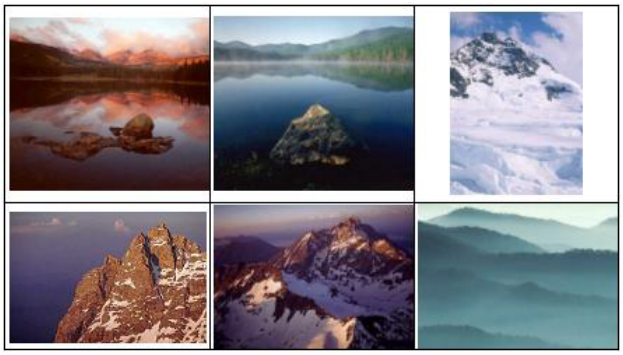


Figure 1. Image examples in cluster “mountain”

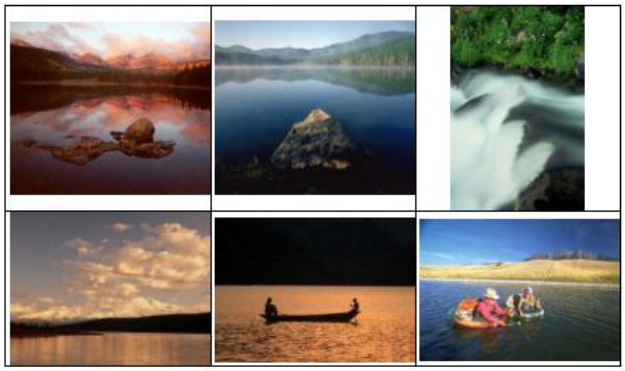


Figure 2. Image examples in cluster “lake”



Figure 3. Image examples in cluster “red flower”



Figure 4. Image examples in cluster “yellow flower”



Figure 5. Image examples in cluster “bouquet”